Adaptive Wind Power Forecasting Using Concept Drift Detection

Complete Project Explanation, Methodology Breakdown & Viva Voce Guide

Project: Final-Year AI/ML Engineering Project

Institution: Panimalar Engineering College (Anna University)

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Target Horizon: 1-Hour-Ahead Generation Forecast

Turbine Rating: 2.0 MW (2,000 kW Aggregated Farm)

1. The Big Picture (What is this project?)

Wind power is one of the most vital clean energy sources, but unlike thermal or hydro plants, its output cannot be dispatched on demand—it depends second-by-second on fluctuating weather. Power grid operators must know **exactly how much electricity the wind farm will produce 1 hour in advance** to balance grid supply and demand, schedule reserves, and avoid expensive fossil-fuel ramp-ups. In this project, we built an intelligent Machine Learning system that consumes meteorological data (wind speed, direction, temperature) and recent generation history to deliver accurate, reliable 1-hour-ahead forecasts for a 2 MW turbine unit.

2. The Core Problem: Concept Drift in Wind Energy

Wind turbine physics follows the non-linear cubic law: $Power \propto (Wind\ Speed)^3$. Because of this cubic relationship, a tiny **10% change in wind speed creates a huge ~33% swing in electrical generation**.
Traditional machine learning models assume *stationarity* (the belief that weather-to-power patterns never change). In the real world, turbines operate outdoors across years where conditions inevitably shift:

- **Turbine Blade Ageing & Derating (Real Drift):** Blade surface erosion, dirt, and mechanical wear degrade aerodynamic efficiency. The same wind produces less power than when the model was trained.
- **New Wake Deficits & Obstacles (Real Drift):** Newly built turbines or physical obstacles create directional wind turbulence and shadow deficits (e.g. in the South-West sector 200°–265°).
- **Sensor Recalibration (Virtual Drift):** Anemometers replaced or recalibrated shift recorded speeds (+4%) without altering the actual physical wind.

When these shifts occur (**Concept Drift**), a standard model trained once (a **Static Model**) fails silently: its error quadruples from 25 kW to 134.7 kW, yet it has no awareness that it has become outdated.

3. The Trap of Blind Calendar Retraining (Why Not Retrain Every Week?)

A common industry workaround is retraining models on a fixed calendar schedule (e.g. every Sunday night). Our empirical research proves this approach has severe flaws:
1. **Late Reaction to Sudden Shifts:** If an obstacle or blade derating occurs on Monday morning, a weekly model remains inaccurate for nearly 6 full days before its next update.
2. **Variance Penalty When Stationary:** When conditions are normal (no drift), retraining on a short 4-week window actually makes forecasts **30.2% worse** (MAE degrades from 25.5 kW to 33.2 kW) because the model forgets rare weather extremes.
3. **Computational Waste:** Performing 80 retrainings when nothing has changed wastes substantial compute and operational effort.

4. The Proposed 4-Step Adaptive Solution

Step 1: Prequential 1-Hour-Ahead Forecast: A HistGradientBoostingRegressor (120 trees, max depth 6) predicts generation using 10 strictly causal features (speed, sin/cos direction, temperature, lagged power t-1, t-2, t-24, rolling power mean 6h, lagged speed, rolling speed mean 3h) with zero future target leakage.

Step 2: Heteroscedastic Error Standardisation: Absolute error naturally scales with power level. We divide the raw error by the expected error for that power level using a 6-quantile calibration map, creating a scale-free score $z(t) = |y(t) - \blacksquare(t)| / m(\blacksquare(t))$ whose baseline mean sits stably at $\mu\blacksquare \approx 1.0$.

Step 3: Sequential Drift Alarm (Page-Hinkley Test): The system tracks the cumulative excess deviation $m(t) - M(t)$. When errors surge, the statistic crosses threshold $\lambda = 15\sigma\blacksquare = 12.0$, sounding an alarm. It detected drift in **21.5 hours on average with zero false alarms** before onset.

Step 4: Smart Adaptive Retraining Controller: An adaptation controller enforces a **48-hour refractory cooldown** (preventing thrashing), retrains on the recent 4-week window (672 h), applies recency decay (half-life 336 h), and downweights data prior to the estimated change point (0.08x).

5. Empirical Benchmark Results (10 Random Seeds)

Evaluation Metric	Static Model (Train Once)	Adaptive Framework (Proposed)	Weekly Retrained (Schedule)	Takeaway / Conclusion
Drift: MAE (kW)	134.7 ± 5.6	**33.7 ± 2.1**	35.3 ± 1.3	**75.0% error reduction** (p = 3e-13)
Drift: RMSE (kW)	178.4 ± 5.6	**53.0 ± 3.7**	62.7 ± 5.4	**70.3% error reduction** (p = 2.2e-4)
Drift: nRMSE (%)	8.92 ± 0.28%	**2.65 ± 0.19%**	3.14 ± 0.27%	Substantial grid reliability improvement
Retrainings Count	0	**16.2 ± 6.6**	80.0	**4.9× fewer retrainings**
Compute Runtime (s)	0.00 s	**1.98 ± 0.82 s**	9.59 ± 0.53 s	**4.9× less computation**
No-Drift MAE (kW)	25.5 ± 0.6	**25.5 ± 0.6**	33.2 ± 1.5	Weekly model suffered **+30.2% penalty**
False Alarms	—	**0 in 10 seeds**	—	Zero false alarms before drift onset

6. Architecture of the 7 Web Dashboard Pages

1. **Overview Dashboard:** Real-time SCADA telemetry, live KPIs, actual vs predicted power chart with drift onset zoom, 80-week MAE divergence curves, and drift alarm timeline.
2. **Forecasting Studio:** Interactive workbench to adjust wind speed, direction, and temperature sliders, compute immediate 1h forecasts, tune hyperparameters, and export results.
3. **Drift Monitoring:** Simulated playback controls (Step 1h, Step 24h, Reset), Page-Hinkley statistic curve vs threshold line $\lambda = 12.0$, Kolmogorov-Smirnov test and PSI input monitors.
4. **Adaptive Learning:** Active model version telemetry, 48h cooldown timer, sample weights curve chart (recency decay + pre-change downweighting), and Manual Retrain trigger.
5. **Model Comparison:** Head-to-head bar charts (Static vs Adaptive vs Weekly), paired Student's t-test and Wilcoxon statistical tables, and automated analytical evaluation.
6. **Dataset Explorer:** Paginated data table, 0-missing-value health verification, empirical wind speed vs power scatter plot with power curve overlay, and 16-sector wind rose.
7. **Experiments & Reports:** Multi-seed test lab (seeds 0–9), sensitivity curves (threshold, cooldown, window, magnitude), and instant downloadable PDF executive report.

7. Three-Sentence Elevator Pitch for Viva Voce Examination

"Our project develops an intelligent 1-hour-ahead wind power forecasting system that detects when turbine degradation or environmental wake effects cause prediction errors to rise. Instead of retraining blindly on a calendar schedule, our system uses scale-standardized residual monitoring and the Page-Hinkley test to retrain only when statistical evidence proves the model has become outdated. This achieves a 75% error reduction under concept drift while requiring 4.9 times less computation than traditional scheduled retraining."